

Stronger Notions of Monotonicity

Chapter 22 - Convex Analysis and Monotone Operator Theory

Lecture Slides

Chapter Overview

- 1 Para, Strict, Uniform, and Strong Monotonicity
- 2 Cyclic Monotonicity
- 3 Rockafellar's Cyclic Monotonicity Theorem
- 4 Monotone Operators on $\mathbb{R}$
- 5 Summary and Relationships

Key Monotonicity Concepts

- **Basic Monotonicity:** An operator $A : \mathcal{H} \rightarrow 2^{\mathcal{H}}$ is monotone if

$$\langle x - y | u - v \rangle \geq 0$$

for all $(x, u), (y, v) \in \text{gra } A$.

- This chapter explores *stronger* notions of monotonicity:
 - 1 Paramonotone
 - 2 Strictly monotone
 - 3 Uniformly monotone
 - 4 Strongly monotone
 - 5 Cyclically monotone
- Each notion imposes additional structure and properties
- Fundamental for understanding optimization algorithms

Paramonotone Operators

Definition

Let K be a real Hilbert space and let $A : \mathcal{H} \rightarrow 2^{\mathcal{H}}$ be an operator. Then A is **paramonotone** if A and A^{-1} are both monotone.

Proposition 22.2

Let K be a real Hilbert space, let $A : \mathcal{H} \rightarrow 2^{\mathcal{H}}$ and $B : \mathcal{K} \rightarrow 2^{\mathcal{K}}$ be paramonotone operators, and let $L \in \mathcal{B}(\mathcal{H}, \mathcal{K})$. Then:

- (i) A^{-1} is paramonotone
- (ii) $A + L^*BL$ is paramonotone

Intuition: Both the operator and its inverse preserve the monotonicity structure.

Strictly Monotone and Strictly Cocoercive Operators

Definition

An operator A is **strictly monotone** if A is monotone and

$$\langle x - y | u - v \rangle = 0, \quad (x, u), (y, v) \in \text{gra } A \implies x = y.$$

Definition

An operator A is **strictly cocoercive** if there exists $\alpha \in]0, 1[$ such that

$$\langle x - y | u - v \rangle \geq \alpha \|u - v\|^2.$$

Example (22.6)

Let D be a nonempty subset of $\mathcal{H}$, let $T : D \rightarrow \mathcal{H}$, let $\alpha \in [-1, 1]$, and set $A = \text{Id} + \alpha T$.

- (i) If T is strictly nonexpansive, then A is strictly monotone.
- (ii) If T is β -Lipschitz continuous with $\beta \in [0, 1]$, then A is strongly monotone with modulus $1 - |\alpha|\beta$.

Uniformly and Strongly Monotone Operators

Definition

An operator A is **uniformly monotone** with modulus $\phi : \mathbb{R}_+ \rightarrow [0, +\infty]$ if

$$\langle x - y | u - v \rangle \geq \phi(\|x - y\|), \quad \forall (x, u), (y, v) \in \text{gra } A.$$

Definition

An operator A is **strongly monotone** with constant $\beta \in \mathbb{R}_{++}$ if

$$\langle x - y | u - v \rangle \geq \beta \|x - y\|^2, \quad \forall (x, u), (y, v) \in \text{gra } A.$$

Hierarchy:

Strongly Monotone $\implies$ Uniformly Monotone $\implies$ Monotone

Monotonicity of Subdifferentials

Example (22.4)

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper and convex. Then:

- (i) ∂f is paramonotone.
- (ii) If f is strictly convex, then ∂f is strictly monotone.
- (iii) If f is uniformly convex with modulus ϕ , then ∂f is uniformly monotone with modulus 2ϕ .
- (iv) If f is strongly convex with constant $\beta \in \mathbb{R}_{++}$, then ∂f is strongly monotone with constant β .

Key insight: Monotonicity properties of ∂f reflect the convexity structure of f .

Existence and Uniqueness

Example (22.12)

Let $A : \mathcal{H} \rightarrow \mathcal{H}$ be monotone and hemicontinuous, and let $r \in \mathcal{H}$. Suppose that one of the following holds:

- (i) A is strictly monotone and $\lim_{\|x\| \rightarrow +\infty} \|Ax\| = +\infty$.
- (ii) A is uniformly monotone with a supercoercive modulus.
- (iii) A is strongly monotone.

Then the equation $Ax = r$ has *exactly one solution*.

Importance: Monotonicity directly implies solvability of linear systems with monotone operators.

Definition of Cyclic Monotonicity

Definition (22.13)

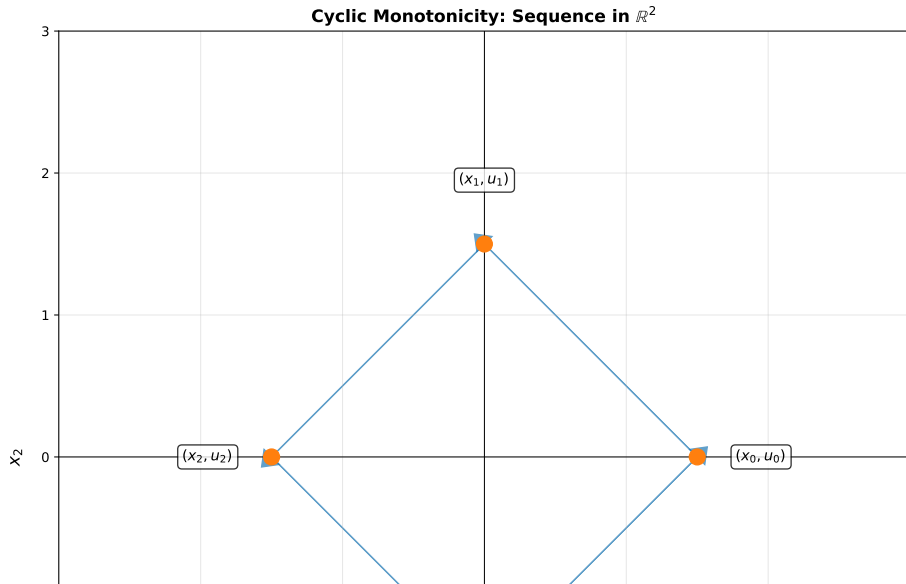
Let $A : \mathcal{H} \rightarrow 2^{\mathcal{H}}$ and let $n \in \mathbb{N}$ be such that $n \geq 2$. Then A is **n -cyclically monotone** if, for every $(x_1, \dots, x_{n+1}) \in \mathcal{H}^{n+1}$ and every $(u_1, \dots, u_n) \in \mathcal{H}^n$,

$$\left. \begin{array}{l} (x_1, u_1) \in \text{gra } A, \\ \vdots \\ (x_n, u_n) \in \text{gra } A, \\ x_{n+1} = x_1 \end{array} \right\} \Rightarrow \sum_{i=1}^n \langle x_{i+1} - x_i | u_i \rangle \leq 0.$$

Remark

- If A is n -cyclically monotone for every $n \geq 2$, then A is **cyclically monotone**.
- **2-cyclic monotonicity** is ordinary monotonicity.

Visual Example of Cyclic Monotonicity



Cyclic Monotonicity of Subdifferentials

Proposition 22.14

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper. Then ∂f is cyclically monotone.

Proof Sketch.

Fix $n \geq 2$. For every $i \in \{1, \dots, n\}$, take $(x_i, u_i) \in \text{gra } \partial f$. By property (16.1) of subdifferentials:

$$\langle x_{i+1} - x_i | u_i \rangle \leq f(x_{i+1}) - f(x_i).$$

Summing over the cycle and using $x_{n+1} = x_1$:

$$\sum_{i=1}^n \langle x_{i+1} - x_i | u_i \rangle \leq \sum_{i=1}^n (f(x_{i+1}) - f(x_i)) = 0.$$



Not All Maximal Monotone Operators are Cyclically Monotone

Example (22.15)

Suppose $\mathcal{H} = \mathbb{R}^2$, and set

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Then A is maximally monotone but **not** 3-cyclically monotone.

Remark

[22.16] The operator A in Example 22.15 is a counterclockwise rotation by $\pi/2$. More generally, a counterclockwise rotation in $\mathbb{R}^2$ by angle $\theta \in [0, \pi/2[$ is n -cyclically monotone if and only if $\theta \in [0, \pi/n]$.

Insight: Cyclic monotonicity is a genuinely stronger condition than maximal monotonicity.

Cyclic Monotonicity and Paramonotonicity

Proposition 22.17

Let $A : \mathcal{H} \rightarrow 2^{\mathcal{H}}$ be 3-cyclically monotone and maximally monotone. Then A is paramonotone.

Proof Sketch.

For points $(x, u), (y, v) \in \text{gra } A$ with $\langle x - y | u - v \rangle = 0$:

By 3-cyclic monotonicity, for any $(z, w) \in \text{gra } A$:

$$0 \geq \langle y - x | u \rangle + \langle z - y | v \rangle + \langle x - z | w \rangle$$

Using the zero inner product and monotonicity arguments, we can show that both (x, u) and (y, u) belong to $\text{gra } A$. □

The Fundamental Result

Theorem (22.18 - Rockafellar)

Let $A : \mathcal{H} \rightarrow 2^{\mathcal{H}}$. Then A is maximally cyclically monotone if and only if there exists $f \in \Gamma_0(\mathcal{H})$ such that $A = \partial f$.

Significance:

- Establishes a *complete characterization* of subdifferentials via cyclic monotonicity
- Every subdifferential is cyclically monotone (Proposition 22.14)
- Conversely, every maximally cyclically monotone operator arises as a subdifferential
- Bridges monotone operator theory and convex analysis

Rockafellar's Theorem - Key Construction

Theorem (22.18 - Rockafellar (continued))

Proof Strategy (converse direction): *Given a maximally cyclically monotone operator A :*

① Define $f : \mathcal{H} \rightarrow [-\infty, +\infty]$ by

$$f(x) := \sup_{n \geq 1} \sup_{(x_1, u_1), \dots, (x_n, u_n) \in \text{gra } A} \left\{ \langle x - x_n | u_n \rangle + \sum_{i=0}^{n-1} \langle x_{i+1} - x_i | u_i \rangle \right\}$$

② Show that $f \in \Gamma_0(\mathcal{H})$ using cyclic monotonicity

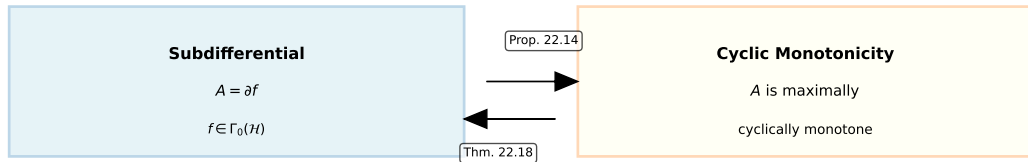
③ Prove that $\text{gra } A \subseteq \text{gra } \partial f$

④ Use maximal cyclic monotonicity to conclude $A = \partial f$

This is a deep result connecting graph properties to convex functions.

Rockafellar's Theorem - Visual Summary

Rockafellar's Cyclic Monotonicity Theorem



Key Consequences

1. Every subdifferential is cyclically monotone
2. A is maximally cyclically monotone $\Leftrightarrow A = \partial f$ for some f
3. Provides structural understanding of monotone operators
4. Connection to optimization: subdifferentials characterize convex functions

Uniqueness of Subdifferentials

Proposition 22.19

Let f and g be functions in $\Gamma_0(\mathcal{H})$ such that $\partial f = \partial g$. Then there exists $\gamma \in \mathbb{R}$ such that $f = g + \gamma$.

Proof Idea.

Consider the special case where f and g are differentiable. If $\partial f = \partial g$, then $\nabla f = \nabla g$ everywhere, which implies $f - g$ is constant.

The general case follows by approximation techniques using $\square g$ and the conjugate operations. □

Consequence: Subdifferentials determine convex functions up to an additive constant.

Characterization on One Dimension

Definition

We introduce a binary relation on $\mathbb{R}^2$ via

$$\forall x_1 = (x_1, u_1) \in \mathbb{R}^2 \quad \forall x_2 = (x_2, u_2) \in \mathbb{R}^2 : \quad x_1 \preceq x_2 \Leftrightarrow x_1 \leq x_2 \text{ and } u_1 \leq u_2.$$

We write $x_1 \prec x_2$ if $x_1 \preceq x_2$ and $x_1 \neq x_2$.

Proposition 22.20

$(\mathbb{R}^2, \preceq)$ is directed and partially ordered, but **not** totally ordered.

Remark: This partial order is crucial for analyzing monotone relations in $\mathbb{R}^2$.

Main Equivalence in One Dimension

Theorem (22.22)

Let $A : \mathbb{R} \rightarrow 2^{\mathbb{R}}$ be such that $\text{gra } A \neq \emptyset$. Then A is monotone if and only if it is **cyclically monotone**.

Corollary (22.23)

Let $A : \mathbb{R} \rightarrow 2^{\mathbb{R}}$ be maximally monotone. Then there exists $f \in \Gamma_0(\mathbb{R})$ such that $A = \partial f$.

Implication: In $\mathbb{R}$, monotone and cyclically monotone are equivalent! Thus Rockafellar's theorem specializes nicely in one dimension.

Intuition for One-Dimensional Case

In $\mathbb{R}$, the structure is particularly rigid:

- The partial order $\preceq$ on $\mathbb{R}^2$ has a special structure
- Any monotone chain in $\mathbb{R}^2$ is automatically cyclically monotone
- This is because we cannot have "loops" in the monotone relation when restricted to one dimension
- Consequently, every maximally monotone operator in $\mathbb{R}$ must arise as a subdifferential

Contrast: In higher dimensions, we can have maximally monotone operators (like rotations) that are not cyclically monotone.

Comparison of Monotonicity Types

Types of Monotonicity for Operators

Monotone

$$\langle x - y | u - v \rangle \geq 0$$

Paramonotone

Both A and A^{-1} monotone

Strictly Monotone

Monotone + inequality strict when $x \neq y$

Uniformly Monotone

$$\langle x - y | u - v \rangle \geq \phi(\|x - y\|)$$

Strongly Monotone

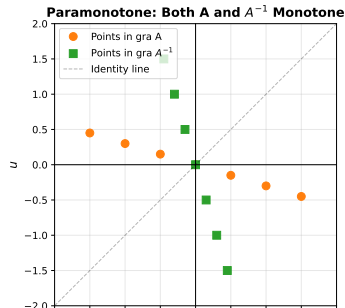
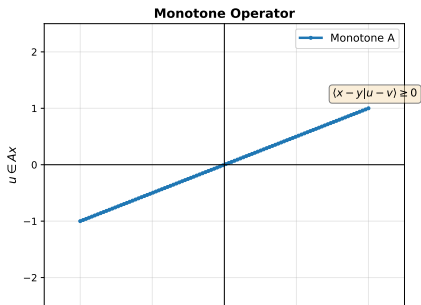
$$\langle x - y | u - v \rangle \geq \beta \|x - y\|^2$$

Cyclically Monotone

$$\sum_{i=1}^n \langle x_{i+1} - x_i | u_i \rangle \leq 0$$

Key Relationships

Property	Definition	Implication
Monotone	$\langle x - y u - v \rangle \geq 0$	Basic concept
Paramonotone	A and A^{-1} both monotone	Invertibility preserved
Strictly Monotone	Monotone + strict when $x \neq y$	Injectivity
Uniformly Monotone	$\langle x - y u - v \rangle \geq \phi(\ x - y\)$	Rate control
Strongly Monotone	$\langle x - y u - v \rangle \geq \beta \ x - y\ ^2$	Quadratic growth
Cyclically Monotone	$\sum_i \langle x_{i+1} - x_i u_i \rangle \leq 0$	Subdifferential



Central Theorems

Subdifferentials are Cyclically Monotone:

If $f \in \Gamma_0(\mathcal{H})$, then

∂f is cyclically monotone

(Proposition 22.14)

Uniqueness of Representation:

If $\partial f = \partial g$, then

$$f = g + \gamma$$

for some $\gamma \in \mathbb{R}$ (Proposition 22.19)

Rockafellar's Theorem:

An operator A is maximally cyclically monotone if and only if

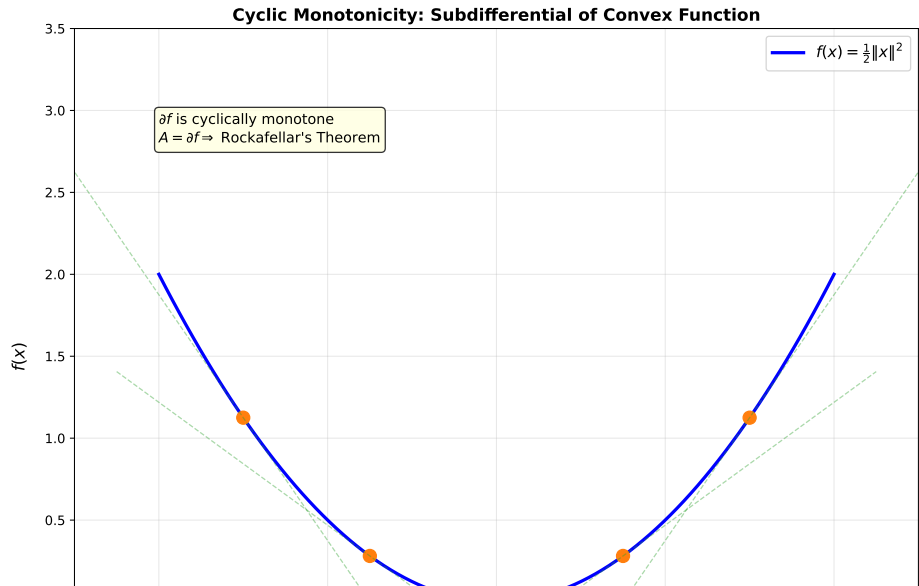
$$A = \partial f$$

for some $f \in \Gamma_0(\mathcal{H})$ (Theorem 22.18)

One-Dimensional Case:

In $\mathbb{R}$, monotone $\Leftrightarrow$ cyclically monotone (Theorem 22.22)

Numerical Examples and Visualizations



Python: Testing Cyclic Monotonicity

```
import numpy as np

def check_cyclic_monotone(graph, n=3):
    """Check if finite cycle satisfies cyclic monotonicity"""
    points = list(graph)
    if len(points) < n:
        return None

    # Test n-cyclic monotonicity
    for i in range(len(points) - n + 1):
        x_vals = [p[0] for p in points[i:i+n]]
        u_vals = [p[1] for p in points[i:i+n]]

        # Compute sum of inner products
        cyclic_sum = 0
        for j in range(n):
            dx = x_vals[(j+1) % n] - x_vals[j]
            cyclic_sum += np.dot(dx, u_vals[j])

        if cyclic_sum > 1e-10: # tolerance
            return False

    return True

# Example: subdifferential of  $f(x) = (1/2)||x||^2$ 
# which is  $\text{partial}_f(x) = \{x\}$ 
graph = [
```

Python: Verifying Operator Properties

```
import numpy as np

def is_monotone(x1, u1, x2, u2):
    """Check monotonicity:  $\langle x_1 - x_2 \mid u_1 - u_2 \rangle \geq 0$ """
    dx = x1 - x2
    du = u1 - u2
    return np.dot(dx, du) >= -1e-10 # tolerance

def is_strongly_monotone(x1, u1, x2, u2, beta):
    """Check strong monotonicity:  $\langle x - y \mid u - v \rangle \geq \beta \|x - y\|^2$ """
    dx = x1 - x2
    du = u1 - u2
    return np.dot(dx, du) >= beta * np.dot(dx, dx) - 1e-10

# Example:  $A(x) = x$  (identity - strongly monotone with  $\beta=1$ )
x1 = np.array([2.0, 1.0])
x2 = np.array([1.0, 3.0])
u1 = x1 #  $A(x_1) = x_1$ 
u2 = x2 #  $A(x_2) = x_2$ 

print(f"Monotone: {is_monotone(x1, u1, x2, u2)}") # True
print(f"Strongly monotone ( $\beta=1$ ): {is_strongly_monotone(x1, u1, x2, u2, 1.0)}") # True
print(f"Strongly monotone ( $\beta=2$ ): {is_strongly_monotone(x1, u1, x2, u2, 2.0)}") # False

# Verification
dx = x1 - x2
du = u1 - u2
```

Example Computations

Example

Let $\mathcal{H} = \mathbb{R}^2$ and $f(x) = \frac{1}{2}\|x\|^2$.

Claim: ∂f is strongly monotone with constant $\beta = 1$.

Verification:

- For all $x, y \in \mathbb{R}^2$: $\partial f(x) = \{x\}$
- Take $(x, u) = (x, x)$ and $(y, v) = (y, y)$ in $\text{gra}(\partial f)$
- Then:

$$\langle x - y | u - v \rangle = \langle x - y | x - y \rangle \tag{1}$$

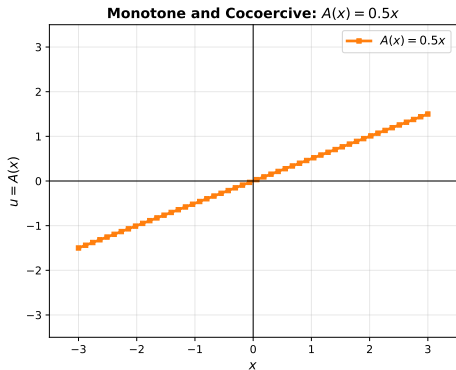
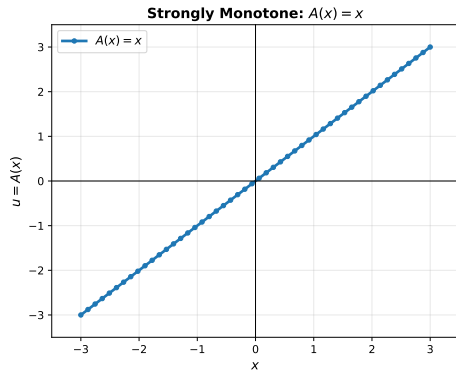
$$= \|x - y\|^2 \geq 1 \cdot \|x - y\|^2 \tag{2}$$

- So ∂f is strongly monotone with $\beta = 1$.

Applications to Optimization

- **Existence and Uniqueness:** Strong monotonicity of ∂f implies the equation $\partial f(x) \ni r$ has a unique solution.
- **Algorithm Design:** Monotonicity properties guide the design of splitting methods (gradient descent, proximal methods, ADMM, etc.).
- **Convergence Analysis:** Different monotonicity constants determine convergence rates of algorithms.
- **Structure Preservation:** Paramonotonicity ensures that compositions and combinations of operators retain nice properties.
- **Inverse Problems:** Monotone operators are fundamental in regularization and ill-posed problem solving.

Further Visualization



Examples of monotone operators: the identity $A(x) = x$ (strongly monotone) and the scaled identity $A(x) = 0.5x$ (monotone and cocoercive).

Key Takeaways

- 1 **Multiple notions of monotonicity** exist, ranging from basic to strongly structured: paramonotone, strictly, uniformly, and strongly monotone.
- 2 **Cyclic monotonicity** is a fundamental concept that characterizes subdifferentials of proper convex functions via Rockafellar's theorem.
- 3 **Rockafellar's Theorem** (22.18) provides a complete characterization: An operator is maximally cyclically monotone $\Leftrightarrow$ it is a subdifferential.
- 4 **Subdifferentials are always cyclically monotone** (Proposition 22.14), while not all monotone operators are cyclically monotone (Example 22.15).
- 5 **In one dimension**, monotone and cyclically monotone are equivalent (Theorem 22.22), making the structure much simpler.
- 6 These concepts are **essential for optimization algorithms** and inverse problem analysis.

References and Further Study

- **Primary Reference:** Bauschke, H.H., and Combettes, P.L. *Convex Analysis and Monotone Operator Theory in Hilbert Spaces*. Springer, 2017. Chapter 22.
- **Key Theorems:**
 - Proposition 22.2: Paramonotonicity preservation
 - Proposition 22.14: Subdifferentials are cyclically monotone
 - Theorem 22.18 (Rockafellar): Cyclic monotonicity characterization
 - Theorem 22.22: One-dimensional equivalence
- **Related Chapters:** Chapter 20 (Monotone Operators), Chapter 21 (Comonotonicity), Chapter 23 (Resolvents).
- **Applications:** Convex optimization, variational inequalities, splitting methods, proximal algorithms.